Euclid's Elements — Book I
Proposition 44

Formal Reconstruction — Technical Documentation

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Applying a parallelogram equal in area to a given triangle

1 Introduction

Proposition I.44 is the first Book I theorem that takes the synthetic content machinery developed in I.35–I.43 and applies it to a prescribed segment. The input consists of a nondegenerate segment AB, a nondegenerate triangle PQR, and a nondegenerate angle XYZ. The output is a parallelogram on the given side AB, with the prescribed angle at B, and with the same content as the given triangle.

The current Lean statement is deliberately synthetic. It does not introduce a numerical area function. The target parallelogram is represented by `hilbertParallelogramTerm`, and equality of content is represented by `HilbertScissorsEquicomplementable`. Thus the theorem constructs points M, L such that

$$ABML \text{ is a parallelogram}, \qquad \angle ABM \cong \angle XYZ,$$

and

$$\triangle PQR \approx ABML,$$

where $\approx$ denotes the project's formal equicomplementability relation.

I.44 is also the first proposition in the application-of-areas block where a small phrase in the classical proof hides a substantial formal obligation. Euclid first invokes I.42 to obtain a parallelogram of the correct content and angle, and then says that one side of this parallelogram is to be placed in a straight line with the given segment. The final Lean proof does not treat this rigid placement as a primitive operation. It is reconstructed from segment construction, angle construction, SAS, SSS, and completion of a parallelogram.

A second hidden obligation occurs later. In the classical diagram two produced lines meet at a point K. Euclid justifies this by an angle argument using I.29 and Postulate 5. The Lean proof reaches the same intersection by a different Euclidean route: if the two carriers were disjoint, they would be parallel; uniqueness and transitivity of parallels would then force two lines through a common point to be parallel, which is impossible. The conclusion is the same, but the formal DAG is not a mechanical transcription of the classical angle-sum proof.

The proof therefore has three sharply separated layers.

- **Geometry**: rigid placement, extensions, intersections, parallel carriers, SameSide/OppositeSide arguments, strict order, and completion of several parallelograms.
- **Content**: a congruent copy preserves scissors content; I.43 identifies the two complements; transitivity then connects the I.42 parallelogram to the final one.
- **Representation**: the I.43 complement terms are sums of triangles, while I.44 exposes whole parallelogram terms. The proof therefore uses alternate triangulations and triangle permutations to bridge the two representations.

The main architectural result of the reconstruction is that I.44 closes with no new geometric axiom and no new content axiom. In particular, the former provisional transport axiom has been eliminated. The public theorem now rests on the previously developed Hilbert and scissors infrastructure together with proposition-local lemmas that expose the geometry suppressed by the classical diagram.

2 The Classical Configuration

Let AB be the given straight segment, let PQR be the given triangle, and let XYZ represent the prescribed rectilineal angle. Euclid first uses I.42 to construct a parallelogram of the required content and angle. In the lettering used by the reconstruction, after rigid placement this auxiliary parallelogram is $BEFG$, with

$$A - B - E \qquad \text{and} \qquad \angle EBG \cong \angle XYZ.$$

The side FG is produced beyond G to H. Through A a line is drawn parallel to GB. The formal order relation is

$$H - G - F, \qquad AH \parallel GB.$$

The carrier HB then meets the carrier FE at K, and the proof recovers the strict relations

$$H - B - K, \qquad F - E - K.$$

Through K a line is drawn parallel to BE. Its intersections with the produced carriers HA and GB are denoted by L and M. The final order data are

$$H - A - L, \qquad G - B - M, \qquad L - M - K.$$

These relations are not decorative labels. They are the exact strict-order facts used by the formal proof.

The resulting large figure is the parallelogram $HLKF$, with diameter HK. The two parallelograms about the diameter are

$$HABG \qquad \text{and} \qquad BMKE.$$

The two complements are precisely the old placed parallelogram $BEFG$ and the target parallelogram $ABML$. This is the configuration to which Proposition I.43 applies.

Figure 1 records the classical Euclidean configuration only. It shows the placed I.42 parallelogram, the produced carriers, the intersection K, the parallel through K, and the final I.43 parallelogram. The stricter order information required by Lean is deliberately reserved for the later formal figures.

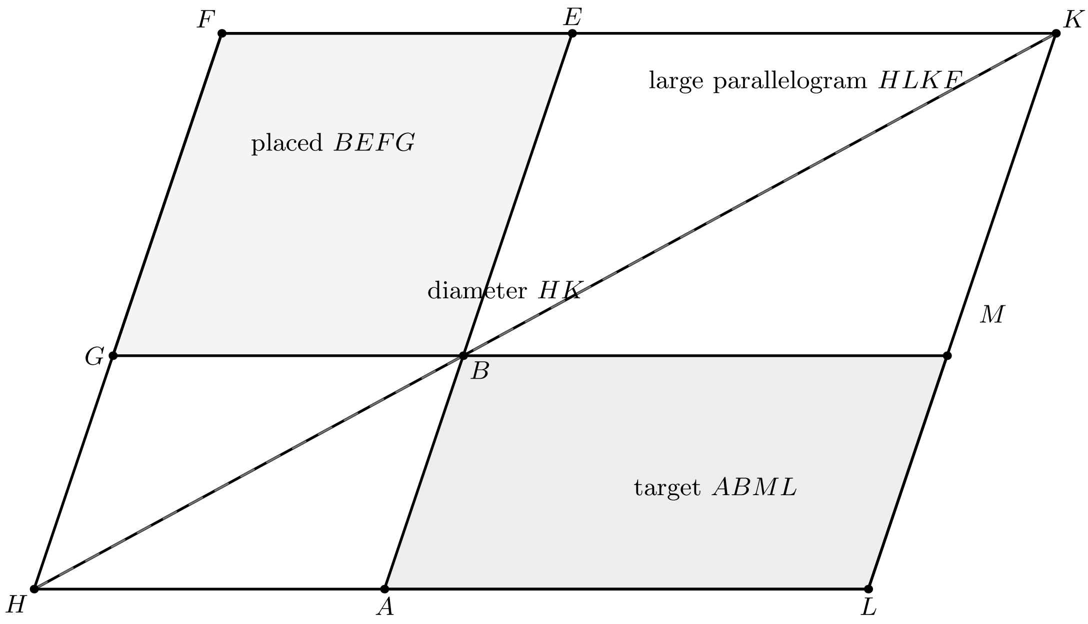

Figure 1: The classical Euclidean configuration. The auxiliary parallelogram $BEFG$ is placed on the produced carrier of AB; the relevant produced lines meet at K; and the parallel through K completes the large parallelogram $HLKF$. The target parallelogram is the complement $ABML$. No additional strict-order claim is encoded by this historical figure.

3 Classical Proof

The classical proof has a compact dependency graph but several diagrammatic steps.

First apply Proposition I.42 to construct a parallelogram $BEFG$ equal to the given triangle and having

$$\angle EBG = \angle XYZ.$$

Place it so that BE is in the same straight line as the prescribed segment AB, with B between A and E. Euclid then produces FG to H and draws AH parallel to BG.

Because $AH \parallel EF$, the transversal HF creates the supplementary angle configuration used in Euclid's text. With I.29 and the angle facts already available from the parallelogram, the interior angles made by the produced lines HB and FE are less than two right angles. Postulate 5 therefore forces the two lines to meet; call their intersection K.

Through K draw a line parallel to EA and produce HA and GB until they meet this new line at L and M. The quadrilateral $HLKF$ is a parallelogram, HK is its diameter, and $HABG$ and $BMKE$ are the two parallelograms about the diameter. Proposition I.43 now gives equality of the complements:

$$ABML = BEFG$$

in Euclid's equality-of-figures sense.

By I.42, $BEFG$ is equal to the given triangle. Common Notion 1 therefore gives the required equality between $ABML$ and the triangle. Finally, $A - B - E$ and $G - B - M$ make $\angle ABM$ and $\angle EBG$ vertical angles, so I.15 gives

$$\angle ABM = \angle EBG = \angle XYZ.$$

Thus the required parallelogram has been applied to AB in the prescribed angle.

The historical dependency graph is therefore

$$\begin{gathered}\triangle PQR,\ AB,\ \angle XYZ \\ \Downarrow\ \text{I.42} \\ BEFG \text{ with the required content and angle} \\ \Downarrow\ \text{place } BE \text{ on the produced carrier of } AB \\ A - B - E \\ \Downarrow\ \text{I.31, I.29, Postulate 5} \\ H - G - F,\ H - B - K,\ F - E - K \\ \Downarrow\ \text{parallel through } K \\ HLKF,\ HABG,\ BMKE \\ \Downarrow\ \text{I.43} \\ ABML = BEFG \\ \Downarrow\ \text{C.N.1 and I.15} \\ ABML = \triangle PQR, \qquad \angle ABM = \angle XYZ.\end{gathered}$$

Two points in this graph are especially important for formalization. The phrase "place BE in a straight line with AB" is not a theorem reference in the classical text, and the meeting of HB and FE is justified through a specifically Euclidean angle argument. Both steps are reconstructed explicitly in Lean.

4 Application of Areas and Rigid Placement

The proposition-specific novelty of I.44 is not the existence of a parallelogram with prescribed content and angle; I.42 already supplies that. The new problem is to put such a parallelogram on a *specified segment* without changing either its shape or its content.

The helper

```
i44_place_parallelogram_shape
```

solves the geometric part of this problem. Suppose $STUV$ is a given parallelogram and $A \neq B$. The proof first extends AB beyond B. On the resulting ray it lays off BE congruent to TS. It then copies the angle STU at B with first arm BE, lays off BG congruent to TU on the constructed ray, and completes BEG to the parallelogram $BEFG$. The output contains

$$A - B - E, \qquad BE \cong TS, \qquad BG \cong TU, \qquad \angle EBG \cong \angle STU.$$

Figure 2 shows only this rigid-placement stage. It is separate from Figure 6 because the two diagrams solve mathematically different problems: the first transports the shape to the prescribed base, while the second performs Euclid's I.43 complement construction.

The companion helper

```
i44_placed_parallelogram_content
```

proves that this rigid copy preserves the scissors content of the original parallelogram. The argument is stronger than a bare appeal to an undefined notion of "same area". It triangulates the two parallelograms and proves congruence of corresponding triangles.

The triangle BEG is congruent to TSU by SAS, since the two sides and their included angle

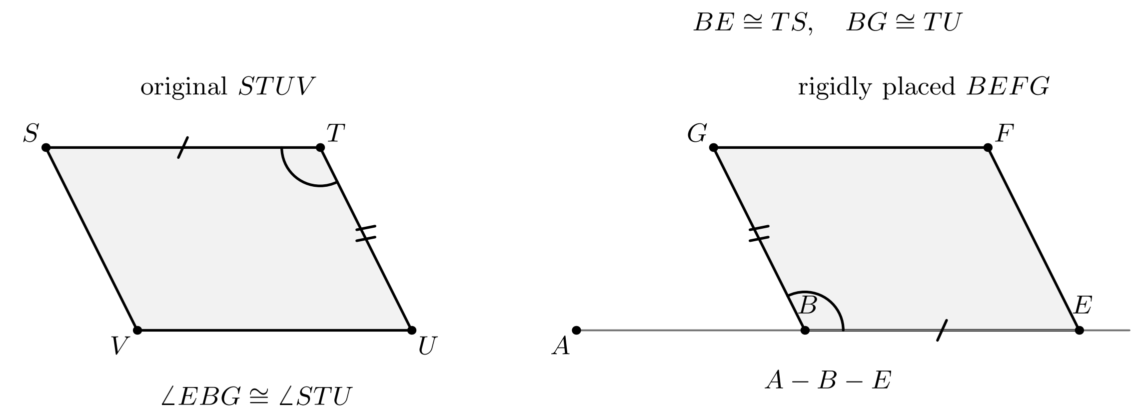

Figure 2: The rigid-placement helper. The original parallelogram $STUV$ is copied to $BEFG$ with $A - B - E$, $BE \cong TS$, $BG \cong TU$, and $\angle EBG \cong \angle STU$. The construction uses segment layoff, angle construction on a chosen side, ray orientation, and completion of a parallelogram. No content theorem is used in the geometric placement itself.

were copied explicitly. This also gives the diagonal congruence

$$EG \cong SU.$$

Opposite-side congruence in the two parallelograms supplies the remaining side relations needed for SSS, yielding

$$\triangle EFG \cong \triangle SVU.$$

The two congruences are converted into `HilbertScissorsEq` relations. The existing helper

```
parallelogram_two_triangulations
```

then identifies their sums with the whole parallelogram terms. Hence the original and placed parallelograms are scissors-equivalent, and therefore equicomplementable.

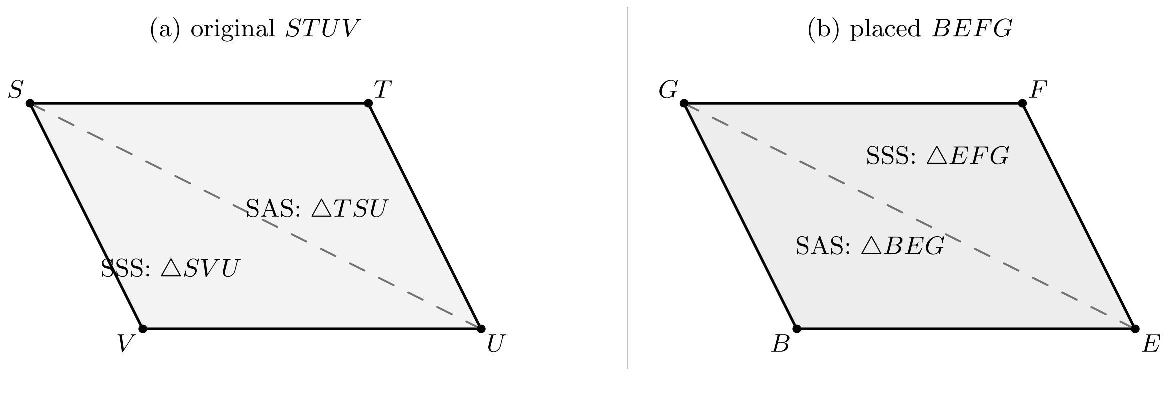

Figure 3: The exact two-piece content certificate used by `i44_placed_parallelogram_content`. The original parallelogram $STUV$ is triangulated along SU and the placed copy $BEFG$ along EG. SAS gives $\triangle BEG \cong \triangle TSU$ and the resulting diagonal congruence supports the SSS comparison $\triangle EFG \cong \triangle SVU$. Adding these two congruent triangle pairs produces the required scissors equality of the parallelograms.

This separation between *shape placement* and *content preservation* is important. The first helper belongs to synthetic geometry; the second is a certificate that the rigid copy has not changed the formal content. Neither helper introduces numerical area.

5 Proof Architecture of the Reconstruction

The final Lean proof can be compressed into seven mathematical stages. The source file contains more helper declarations, but many of them are local normalization lemmas for order or carrier orientation. The stages below are the mathematically significant transitions.

5.1 Step 1: obtain a content-and-angle parallelogram from I.42

The public theorem begins by calling

```
euclid_proposition_42
```

on the triangle PQR and the angle XYZ. It obtains points E, F, G such that $FERG$ is a parallelogram,

$$\angle FER \cong \angle XYZ,$$

and

$$\triangle PQR \approx FERG.$$

At this point the parallelogram has the correct content and angle but is not located on the prescribed segment AB.

5.2 Step 2: rigidly place the I.42 parallelogram beyond B

The theorem `i44_transport_onto_line` treats the I.42 output as a general parallelogram $STUV$. The helper `i44_place_parallelogram_shape` constructs the placed copy $BEFG$ with $A-B-E$ and with two adjacent sides and the included angle copied from $STUV$.

This is the formal replacement for the classical phrase that the side of the I.42 parallelogram is to be placed in the same straight line as the given segment. In an earlier provisional architecture this step was packaged as a local transport axiom. The current file derives it completely.

5.3 Step 3: certify the content of the rigid copy

The helper `i44_placed_parallelogram_content` proves

$$STUV \approx BEFG.$$

Its proof uses SAS, opposite-side congruence of parallelograms, SSS, and the two triangulations of a parallelogram. Thus the content transport is reduced to finite congruent-piece geometry rather than postulated as an area law.

5.4 Step 4: construct the positioned Euclid configuration

Starting with $A - B - E$ and the placed parallelogram $BEFG$, the helper

```
i44_positioned_content
```

constructs the full I.43 configuration. The first key result is

```
i44_construct_H_ordered
```

which returns H with
$$H - G - F, \qquad AH \parallel GB.$$

The incidence part first constructs the parallel through A and proves that it meets the carrier FG. The order part then uses line separation to show that the intersection lies beyond G rather than in another collinear position.

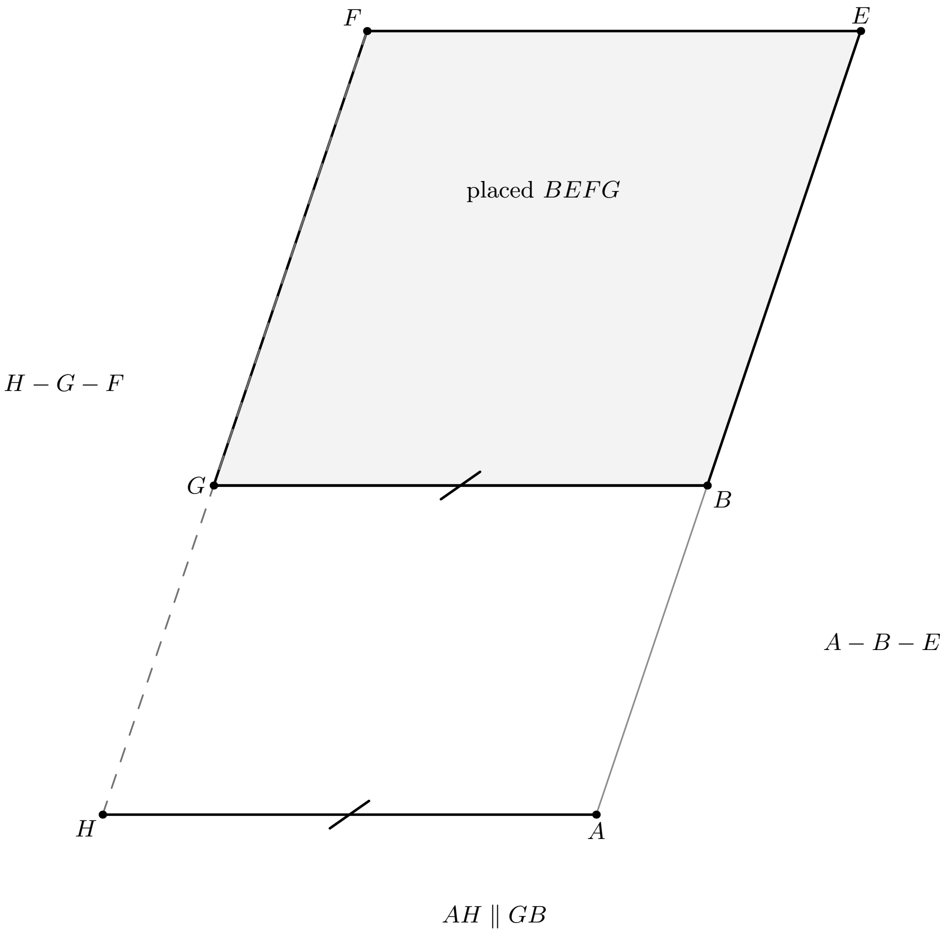

Figure 4: The first new formal state after rigid placement. The carrier FG is produced to an intersection H with the parallel through A, and the proof establishes the strict relation $H - G - F$ together with $AH \parallel GB$. The position $H - G - F$ is proved by line separation; it is not read from the drawing.

The next key result is

```
i44_construct_K_ordered
```

which returns K with
$$H - B - K, \qquad F - E - K.$$

This is where the formal proof departs most visibly from Euclid's literal argument. Instead of reproducing the sum-of-interior-angles calculation, it supposes that the carriers HB and EF are disjoint. Disjointness gives $HB \parallel EF$. But the old parallelogram already gives $GB \parallel EF$. Euclidean transitivity/uniqueness of parallels would then force HB and GB to be parallel, impossible because they share B. Hence the carriers meet. SameSide/OppositeSide arguments then identify the correct strict order on both lines.

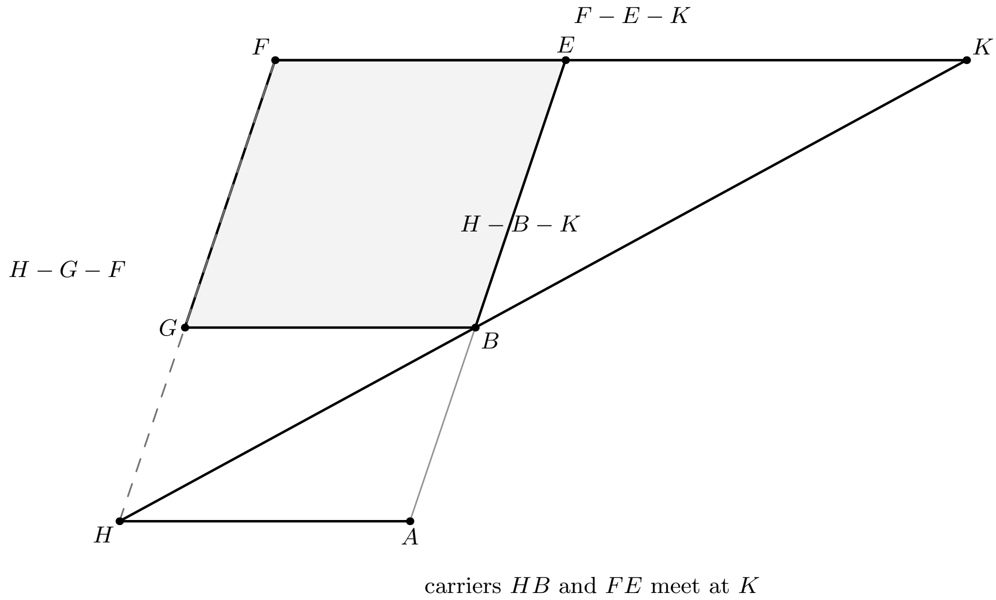

Figure 5: The next geometric state. The produced carriers HB and FE meet at K, and the formal proof recovers both strict orders $H - B - K$ and $F - E - K$. Lean obtains the existence of this intersection by a Euclidean uniqueness/transitivity-of-parallels contradiction rather than by replaying Euclid's literal angle-sum argument.

Finally `i44_construct_LM` draws the carrier through K parallel to BE and finds its intersections with the carriers HA and GB. The order helpers recover

$$H - A - L, \qquad L - M - K, \qquad G - B - M.$$

The last relation is obtained by a separate line-separation argument in `i44_order_GBM` and is later essential for the vertical-angle step.

5.5 Step 5: recognize the four parallelograms needed by I.43

The construction is then repackaged into the four parallelogram statements that express the classical diagram:

$HABG$	first parallelogram about the diameter,
$BMKE$	second parallelogram about the diameter,
$HLKF$	large parallelogram,
$ABML$	target parallelogram.

These conclusions are supplied by the helpers

```
i44_first_diagonal_parallelogram
i44_second_diagonal_parallelogram
i44_big_parallelogram
i44_target_parallelogram
```

The proofs are mainly parallel-carrier transport and recognition of the `IsParallelogram` definition.

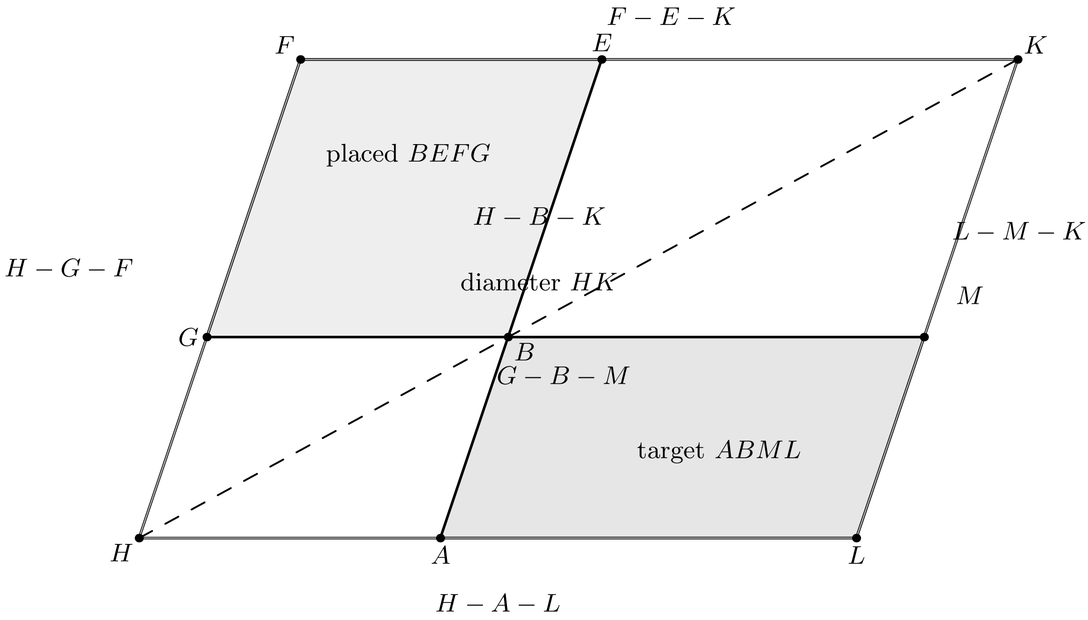

Figure 6: The complete formal configuration immediately before Proposition I.43 is invoked. The large parallelogram is $HLKF$ with diameter HK; the parallelograms about the diameter are $HABG$ and $BMKE$; and the complements are the placed parallelogram $BEFG$ and the target $ABML$. The displayed strict relations $H - A - L$, $H - G - F$, $H - B - K$, $F - E - K$, $G - B - M$, and $L - M - K$ are exactly the order data established by the construction and order helpers.

5.6 Step 6: use I.43 and transfer the angle

The helper

```
i44_i43_content_bridge
```

feeds the constructed data to `euclid_proposition_43`. The point mapping is

I.43 name	A	B	C	D	K	G	E	F	H
I.44 point	H	L	K	F	B	A	G	M	$E.$

Thus the I.43 large parallelogram is $HLKF$, its diameter point is B, and the two parallelograms about the diameter are $HABG$ and $BMKE$.

The public I.43 theorem returns its complements as sums of triangle terms, whereas I.44 wants whole parallelogram terms. The bridge uses

```
parallelogram_two_triangulations
```

together with triangle permutation identities to prove

$$ABML \approx BEFG.$$

This is a representation bridge, not a new content axiom.

For the angle, $A - B - E$ and $G - B - M$ expose vertical angles. The helper

```
i44_angle_transfer
```

first proves that ABM is nondegenerate from the target parallelogram, then invokes the existing vertical-angle theorem and angle transitivity to obtain

$$\angle ABM \cong \angle EBG \cong \angle XYZ.$$

5.7 Step 7: chain the two content transports

The derived transport theorem now has

$$STUV \approx BEFG \qquad \text{and} \qquad BEFG \approx ABML.$$

Transitivity gives

$$STUV \approx ABML.$$

At the top level, I.42 already supplied

$$\triangle PQR \approx STUV,$$

so one last transitivity step proves

$$\triangle PQR \approx ABML.$$

The actual formal DAG is therefore

$$\begin{gathered}
PQR,\ AB,\ XYZ \\
\Downarrow\ \text{I.42} \\
STUV \text{ with prescribed content and angle} \\
\Downarrow\ \text{segment and angle construction} \\
A-B-E,\ BEFG \text{ rigid copy of } STUV \\
\Downarrow\ \text{SAS + SSS + parallelogram triangulations} \\
STUV \approx BEFG \\
\Downarrow\ \text{parallel/intersection/order construction} \\
H-G-F,\ H-B-K,\ F-E-K,\ H-A-L,\ L-M-K,\ G-B-M \\
\Downarrow \\
HLKF,\ HABG,\ BMKE,\ ABML \text{ parallelograms} \\
\Downarrow\ \text{I.43 + representation bridge} \\
ABML \approx BEFG \\
\Downarrow\ \text{I.15 / VerticalAngles} \\
\angle ABM \cong \angle XYZ \\
\Downarrow\ \text{content transitivity} \\
\triangle PQR \approx ABML.
\end{gathered}$$

6 Lean Formalization

The current file imports exactly the two immediately preceding construction and complement results used by the public theorem:

```
import CGJteamLab.Proposition42
import CGJteamLab.Proposition43
```

The public theorem, written in the project's documentation-friendly ASCII notation, is

```
theorem euclid_proposition_44
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B P Q R X Y Z : Geo.Point)
    (hAB : Not (A = B))
    (hPQR : Not (Collinear Geo P Q R))
    (hXYZ : Not (Collinear Geo X Y Z)) :
    exists M L : Geo.Point,
      IsParallelogram Geo A B M L /\
      Geo.AngleCongruent A B M X Y Z /\
      HilbertScissorsEquicomplementable Geo
        (hilbertScissorsTriangle Geo P Q R)
        (hilbertParallelogramTerm Geo A B M L) := by
  -- proof in Proposition44.lean
```

The decisive derived transport theorem has the following shape:

```
theorem i44_transport_onto_line
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B S T U V X Y Z : Geo.Point)
    (hAB : Not (A = B))
    (hParallelogram : IsParallelogram Geo S T U V)
    (hAngle : Geo.AngleCongruent S T U X Y Z) :
    exists M L : Geo.Point,
      IsParallelogram Geo A B M L /\
      Geo.AngleCongruent A B M X Y Z /\
      HilbertScissorsEquicomplementable Geo
        (hilbertParallelogramTerm Geo S T U V)
        (hilbertParallelogramTerm Geo A B M L) := by
  -- rigid placement + positioned I.43 construction
```

This theorem is the formal content of the phrase "apply the parallelogram to the given line". It is no longer an axiom.

The two helpers that isolate rigid placement and its content certificate are

```
theorem i44_place_parallelogram_shape
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B S T U V : Geo.Point)
    (hAB : Not (A = B))
    (hParallelogram : IsParallelogram Geo S T U V) :
    exists E F G : Geo.Point,
      Geo.Between A B E /\
      IsParallelogram Geo B E F G /\
      Geo.AngleCongruent E B G S T U /\
      Geo.Congruent B E T S /\
      Geo.Congruent B G T U := by
  -- segment layoff, angle copy, ray control, fourth vertex
```

and

```
theorem i44_placed_parallelogram_content
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (B E F G S T U V : Geo.Point)
```

```
    (hOriginal : IsParallelogram Geo S T U V)
    (hPlaced : IsParallelogram Geo B E F G)
    (hAngle : Geo.AngleCongruent E B G S T U)
    (hBE_TS : Geo.Congruent B E T S)
    (hBG_TU : Geo.Congruent B G T U) :
    HilbertScissorsEquicomplementable Geo
      (hilbertParallelogramTerm Geo S T U V)
      (hilbertParallelogramTerm Geo B E F G) := by
  -- SAS + SSS + the two parallelogram triangulations
```

The positioned core is

```
theorem i44_positioned_content
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B E F G : Geo.Point)
    (hABE : Geo.Between A B E)
    (hPar : IsParallelogram Geo B E F G) :
    exists M L : Geo.Point,
      IsParallelogram Geo A B M L /\
      Geo.Between G B M /\
      HilbertScissorsEquicomplementable Geo
        (hilbertParallelogramTerm Geo A B M L)
        (hilbertParallelogramTerm Geo B E F G) := by
  -- H, K, L, M construction + I.43
```

The mathematically significant local declarations can be grouped by role.

- `i44_place_parallelogram_shape`: rigid placement on the prescribed base;
- `i44_placed_parallelogram_content`: congruent-piece certificate for the rigid copy;
- `i44_construct_H_ordered`: construct $H - G - F$ with $AH \parallel GB$;
- `i44_construct_K_ordered`: construct $H - B - K$ and $F - E - K$;
- `i44_construct_LM`: construct the parallel through K and its two intersections;
- the three order helpers for $H - A - L$, $L - M - K$, and $G - B - M$: recover strict order from carrier data;
- `i44_i43_content_bridge`: adapt I.43 to whole parallelogram terms;
- `i44_positioned_content`: assemble the full positioned construction;
- `i44_angle_transfer`: transfer the prescribed angle by vertical angles;
- `i44_transport_onto_line`: package the reusable high-level transport theorem.

The remaining proposition-local helpers mostly recognize parallelograms, transport parallelism along collinear carriers, or normalize line orientation. They are essential to compilation but do not represent independent new mathematical principles.

7 Relationship with Earlier Propositions

7.1 Proposition I.15

The last angle step is exactly the vertical-angle theorem. Once the order lemmas have established

$$A - B - E \qquad \text{and} \qquad G - B - M,$$

I.15 identifies $\angle ABM$ with $\angle EBG$. In Lean this appears through the already exposed helper `VerticalAngles`; no new angle theorem is proved in I.44.

7.2 Proposition I.31 and the Euclidean parallel layer

Classically, I.31 is invoked to draw the parallels through A and K. The Lean file does not call the historical proposition by name. It uses the Hilbert-level construction

```
hilbert_parallel_through_point_exists
```

and the developed parallel-carrier infrastructure.

More importantly, the existence of the required intersections is proved by Euclidean uniqueness/transitivity of parallels. Thus the formal proof has a clear Euclidean dependency even where it does not replay the literal I.29 plus Postulate-5 angle argument.

7.3 Proposition I.42

I.42 is the first direct Book I dependency of the final theorem. It converts the given triangle and angle into a parallelogram with the desired content and angle. I.44 adds the missing location constraint: the parallelogram must be applied to a specified segment.

The final architecture therefore treats I.42 as a content-and-shape producer, not as a theorem that already knows where the resulting parallelogram is to be placed.

7.4 Proposition I.43

I.43 is the central content theorem inside the positioned construction. Once $HLKF$, $HABG$, and $BMKE$ have been recognized, I.43 states that the two remaining complements have equal synthetic content. The local bridge `i44_i43_content_bridge` does not strengthen I.43. It only converts its triangle-sum representation of the complements into the parallelogram terms $ABML$ and $BEFG$ used by I.44.

7.5 Propositions I.34–I.41

These propositions are not replayed directly by the public I.44 theorem, but they remain present through its two major dependencies. I.42 already rests on the forward content theory of I.37–I.41, and I.43 rests on the diagonal parallelogram theorem I.34 together with the scissors calculus. I.44 is therefore the first point where those two branches are combined in an active construction.

7.6 Proposition I.45

I.45 generalizes the application-of-areas idea from a triangle to an arbitrary rectilineal figure. The transport theorem isolated in I.44 is therefore a natural reusable unit for the next stage: once a suitable parallelogram of the required total content is available, it can be applied to a prescribed segment with a prescribed angle.

8 Hilbert and Book Zero Components Used

8.1 Betweenness, extensions, and carrier incidence

The formal proof uses strict betweenness much more extensively than the classical text writes. The core order relations are

$$A - B - E, \quad H - G - F, \quad H - B - K, \quad F - E - K, \quad H - A - L, \quad L - M - K, \quad G - B - M.$$

They are obtained by segment extension, line intersection, and plane-separation arguments. Carrier identification repeatedly uses incidence uniqueness for two distinct points.

The proof also makes heavy use of the bridge between a Hilbert incidence line and the extensional `Geo.PointLine` representation. This is what lets the code move safely between line-uniqueness arguments and the strict `Geo.Parallel` relation.

8.2 SameSide and OppositeSide

SameSide/OppositeSide data are the main device for turning a bare line intersection into the correct order statement. For example, `i44_construct_H_ordered` uses the line GB as a separator to prove that the intersection on FG lies beyond G. The K, L, and M order helpers use analogous arguments.

The proof of $G - B - M$ is particularly revealing. It shows that H and G are on the same side of BE, while K and M are on the same side of BE. Since $H - B - K$, the points H and K lie on opposite sides. Transporting through the two SameSide relations gives G and M on opposite sides, and uniqueness of the intersection with BE identifies the crossing point with B.

These side-of-line facts are invisible in the short Euclidean proof but are central to the formal reconstruction.

8.3 Segment construction, SameRay, and angle construction

Rigid placement uses Hilbert segment construction twice: once to copy the first side of the original parallelogram onto the extension of AB, and once to copy the second side onto the constructed angle ray. The first copy must remain on the ray beyond B, and the second must remain on the copied angle ray. SameRay machinery records these orientation facts.

Angle construction is used with an explicit carrier and a chosen side. This produces more information than the classical phrase “construct an equal angle”, and that information is exactly what is needed to prove that the new point G is off the line BE and that BEG is a genuine triangle.

8.4 Euclidean parallelism

The theorem is genuinely Euclidean. The class assumption is

```
[HilbertEuclideanPlane Geo]
```

and the Euclidean input is used in several places:

- construction and uniqueness of a parallel through a point;
- transitivity of distinct parallel carriers;
- proving that the constructed parallel through A meets FG;
- proving that HB meets EF;
- identifying carriers that are both parallel to the same line and share a point;
- completing and recognizing the several parallelograms in the I.43 configuration.

This is the formal counterpart of Euclid's use of I.29 and Postulate 5. In neutral geometry the relevant uniqueness/intersection conclusions are not available in this form.

8.5 Congruence and parallelogram infrastructure

The rigid-copy content proof uses SAS and SSS. It also uses

```
ParallelogramOppositeSidesCongruent
hilbert_parallelogram_fourth_vertex_exists
```

Together they show that a shape copied by two adjacent sides and the included angle is congruent piecewise after triangulation, and that the copied three vertices determine a genuine parallelogram.

No new congruence axiom or parallelogram axiom is introduced in I.44.

8.6 Scissors content infrastructure

The content layer uses only the existing scissors operations, principally

```
scissors_congruent
HilbertScissorsEq.add
HilbertScissorsEq.trans
equicomplementable_of_scissorsEq
equicomplementable_symm
equicomplementable_trans
equicomplementable_transport
parallelogram_two_triangulations
```

I.44 neither defines numerical area nor introduces subtraction or a content order. The only equality-of-figures conclusion is the already established synthetic relation of equicomplementability.

9 Source Audit

9.1 Euclid, Heath, and Joyce

The classical sources agree on the main construction. I.42 supplies a parallelogram equal to the given triangle in the prescribed angle. One side is placed on the produced carrier of the given segment. The side FG is produced, a parallel through A is drawn, and the two produced lines are forced to meet by the Fifth Postulate. A parallel through the intersection creates the large parallelogram and the two parallelograms about its diameter. I.43 identifies the complements, and I.15 supplies the final vertical-angle identity.

These sources control the historical statement and the classical DAG. They also expose the two formal pressure points: the unreferenced placement step and the specifically Euclidean meeting step.

9.2 Wilkins

Wilkins gives the same dependency pattern with explicit attention to the produced lines and the I.43 configuration. His account is useful as a control on point order and on which quadrilaterals are intended to be the two complements. The current Lean diagram follows those order constraints rather than relying on a visually plausible but logically weaker collinearity sketch.

9.3 Hilbert and Borsuk–Szmielew

Hilbert's Chapter IV and the Borsuk–Szmielew treatment control the semantics of content. Their plane-area theory distinguishes equidecomposability from equicomplementability and develops the forward parallelogram/triangle content theorems without requiring the continuity group. In particular, the non-Archimedean distinction warns against replacing the current target by an unjustified global equidecomposability statement.

The same sources reorganize the geometry away from Euclid's proposition numbers. Segment layoff, angle construction, line order, and parallel uniqueness are reusable parts of the axiomatic geometry. This is why the Lean proof can derive the placement and intersection steps without introducing a new I.44-specific axiom.

9.4 Hartshorne

Hartshorne places I.35–I.48 in the area part of Book I and describes I.44 as the construction of a parallelogram with a given side and angle equal to a given triangle. His discussion of "equal content" is also the correct semantic control for the present documentation: Euclid has not introduced a real-valued area function, so equality of figures in this block should not be silently rewritten as a numerical formula.

Hartshorne's analysis of the parallel postulate also supports the formal classification. The Fifth Postulate and Playfair uniqueness are equivalent over the developed neutral theory. The Lean proof chooses the uniqueness form of the Euclidean input when proving the hidden intersections.

9.5 Greenberg

Greenberg supplies an independent modern control on the same Euclidean boundary: Euclid's Fifth Postulate and Hilbert/Playfair parallel uniqueness are equivalent in the neutral setting. This explains why the formal replacement of Euclid's less-than-two-right-angles argument by a uniqueness-of-parallels contradiction does not change the axiomatic strength of the proposition.

9.6 Beeson proof corpus

The Beeson/Tarski corpus remains useful in the project as a gap audit for classical constructions, especially where a traditional diagram names an intersection without isolating its existence or orientation. The final I.44 proof, however, is not organized around Tarski tactics. Its actual DAG is the Hilbert-style one documented above: explicit line carriers, SameSide/OppositeSide order recovery, Euclidean parallel uniqueness, and the project's own scissors representation.

10 Formalization Notes

10.1 Geometry

The geometric cost of I.44 is substantially higher than the length of the classical text suggests. The formal proof must establish all of the following, rather than reading them from a picture:

- the first copied side lies beyond B, giving $A - B - E$;
- the copied angle ray produces a point genuinely off the carrier BE;
- the fourth vertex completing the placed parallelogram exists;
- the parallel through A actually meets the produced carrier FG;
- that intersection is ordered as $H - G - F$;
- the carriers HB and EF actually meet;
- their intersection satisfies both $H - B - K$ and $F - E - K$;
- the parallel through K meets the two required carriers;
- those intersections satisfy $H - A - L$, $L - M - K$, and $G - B - M$;
- the four quadrilaterals used by I.43 satisfy the strict `IsParallelogram` relation.

No continuity axiom is used to obtain these intersections. Their existence comes from the developed Euclidean parallel theory plus incidence and plane-separation arguments.

10.2 Content

There are two content transports and they have different proofs.

First, the arbitrary I.42 parallelogram is rigidly copied to $BEFG$. Here the proof actually produces a direct scissors equality by congruent triangulation: SAS for one triangle, SSS for the other, and addition of the two congruent pieces.

Second, $BEFG$ is transformed into the target $ABML$ by the I.43 complement construction. Here the conclusion is equicomplementability, exactly matching the public content strength of I.43. The final theorem composes these relations by transitivity.

At no stage is equal content used to infer an incidence, order, or metric fact. All geometry is constructed first; content is then transported forward. Consequently I.44 needs neither the positivity principle used at the converse boundary I.39 nor any numerical area theorem.

10.3 Representation

The public orientation of the target is

```
IsParallelogram Geo A B M L
Geo.AngleCongruent A B M X Y Z
```

This orientation matters. The prescribed angle is the angle at B between the actual adjacent sides BA and BM. It should not be represented using a diagonal arm.

The I.43 bridge has a second representation issue. I.43 returns complements as explicit sums of two triangle terms, while I.44 wants a whole parallelogram term. The helper

```
i44_i43_content_bridge
```

uses alternate triangulations and triangle permutation identities to prove that these are the same formal scissors objects up to `HilbertScissorsEq`. No new polygon-region semantics is introduced.

The formal figures therefore represent finite triangle sums and parallelogram terms, not geometric interiors with a measure attached to them.

10.4 Axiomatic status

Proposition I.44 is Euclidean, not neutral. The explicit type-class context is

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

Hilbert Group IV (the Euclidean parallel axiom) is therefore used. Its formal role is uniqueness/transitivity of parallels and the intersection arguments described above. Parallel existence is provided by the developed construction layer; the specifically Euclidean strength is needed when the proof excludes a second distinct parallel through the same point. This has the same axiomatic strength as the classical use of Postulate 5, although the formal proof uses a different equivalent route.

Group III congruence construction is also essential: segment layoff, angle construction, SAS, SSS, and opposite-side congruence of parallelograms are used in the rigid-placement certificate. No Group V continuity principle is used.

The current proposition introduces no active local axiom. In particular, the provisional `i44_transport_onto_line` axiom has been replaced by the derived theorem of the same mathematical purpose. No new theorem is added to `HilbertInterface`, `HilbertBookZero`, or `HilbertScissors` by I.44 itself.

One source-level comment at the beginning of the production file still describes steps 2–6 as being packaged by a local axiom. That comment is historical and is now stale: the active declaration later in the same file is the derived theorem `i44_transport_onto_line`, while the former axiom declaration survives only inside a block comment. This is a documentary discrepancy, not an active foundational dependency.

11 Dependency Summary

The formal dependency chain is

$$\begin{gathered}
\text{Hilbert incidence, order, congruence, and Euclidean parallel axioms} \\
\Downarrow \\
\text{segment/angle construction} + \text{SameRay} + \text{SameSide/OppositeSide} \\
\Downarrow \\
\text{parallel uniqueness/transitivity} + \text{parallelogram infrastructure} \\
\Downarrow \\
\text{Proposition I.42: content-and-angle parallelogram} \\
\Downarrow \\
\texttt{i44_place_parallelogram_shape} \\
\Downarrow \\
\texttt{i44_placed_parallelogram_content} \\
\Downarrow \\
\text{ordered construction of } H, K, L, M \\
\Downarrow \\
\text{large and diagonal parallelogram configuration} \\
\Downarrow \\
\text{Proposition I.43} + \texttt{i44_i43_content_bridge} \\
\Downarrow \\
\texttt{i44_positioned_content} \\
\Downarrow \\
\text{I.15 / vertical-angle transfer} + \text{content transitivity} \\
\Downarrow \\
\texttt{i44_transport_onto_line} \\
\Downarrow \\
\texttt{euclid_proposition_44}.
\end{gathered}$$

The classical and formal DAGs may be summarized as

$$\begin{aligned}
\text{Euclid: I.42} &\;\to\; \text{placement} \;\to\; \text{I.31/I.29/Post.5} \\
&\;\to\; \text{I.43} \;\to\; \text{I.15}, \\
\text{Lean: I.42} &\;\to\; \text{derived rigid placement} \\
&\;\to\; \text{Euclidean parallel/intersection order} \\
&\;\to\; \text{I.43 bridge} \;\to\; \text{VerticalAngles}.
\end{aligned}$$

New geometric axiom:	no
New content axiom:	no
Provisional I.44 transport axiom retained as active axiom:	no
New proposition-local helpers:	yes
New reusable high-level transport helper:	yes
New Book Zero theorem:	no
New Interface/Scissors theorem:	no
Numerical area introduced:	no
Hilbert Group IV used:	yes
Continuity used:	no
Public `Proposition44.lean` compiles:	yes
Full `lake build` re-run in this documentation pass:	no
GitHub CI status checks on the committed code:	none reported
`sorry` introduced in `Proposition44.lean`:	no

Diagrammatic audit. The revised documentation uses six figures, each tied to a genuinely new geometric state. Figure 1 is reserved for the classical Euclidean construction. Figure 2 isolates the derived rigid placement on the prescribed base, while Figure 3 records the distinct two-triangle scissors certificate proving that the rigid copy preserves content. Figures 4 and 5 expose the two ordered intersection stages that the classical proof compresses into produced-line language. Figure 6 is used only after L and M have been constructed, when the complete ordered I.43 configuration first exists. No separate figure is added for the four parallelogram-recognition helpers, the I.43 representation bridge, the vertical-angle theorem, or the final transitivity steps: those operations introduce no new points, lines, intersections, or regions beyond the already displayed configurations. In particular, the figures record geometry rather than theorem-call or normalization state.

12 Conclusion

Proposition I.44 is a major architectural checkpoint in the Book I reconstruction. The classical proof combines I.42 and I.43 with a short parallel construction and a vertical-angle observation. The formal proof shows that the apparent simplicity hides two genuine tasks: a rigid placement of the I.42 parallelogram on a prescribed segment, and a large amount of order information needed to instantiate I.43 correctly.

Both tasks are now derived rather than assumed. The rigid placement is built from segment and angle construction and certified by SAS/SSS triangulation. The positioned construction derives H, K, L, M, recovers the strict order relations by plane separation, recognizes the four required parallelograms, and then invokes I.43 through a representation bridge. Vertical angles close the prescribed-angle condition, and equicomplementability transitivity closes the content condition.

The proposition therefore introduces no new foundational debt. It remains fully synthetic: no numerical area, no polygon measure, no continuity axiom, and no application-of-area transport axiom. At the same time it produces a useful high-level theorem, `i44_transport_onto_line`, which packages a substantial geometric construction into a form suitable for later Book I work. This is precisely the kind of compression the Hilbert/Book Zero architecture is intended to provide: the final theorem is short because the hidden geometry has been made explicit and proved once.